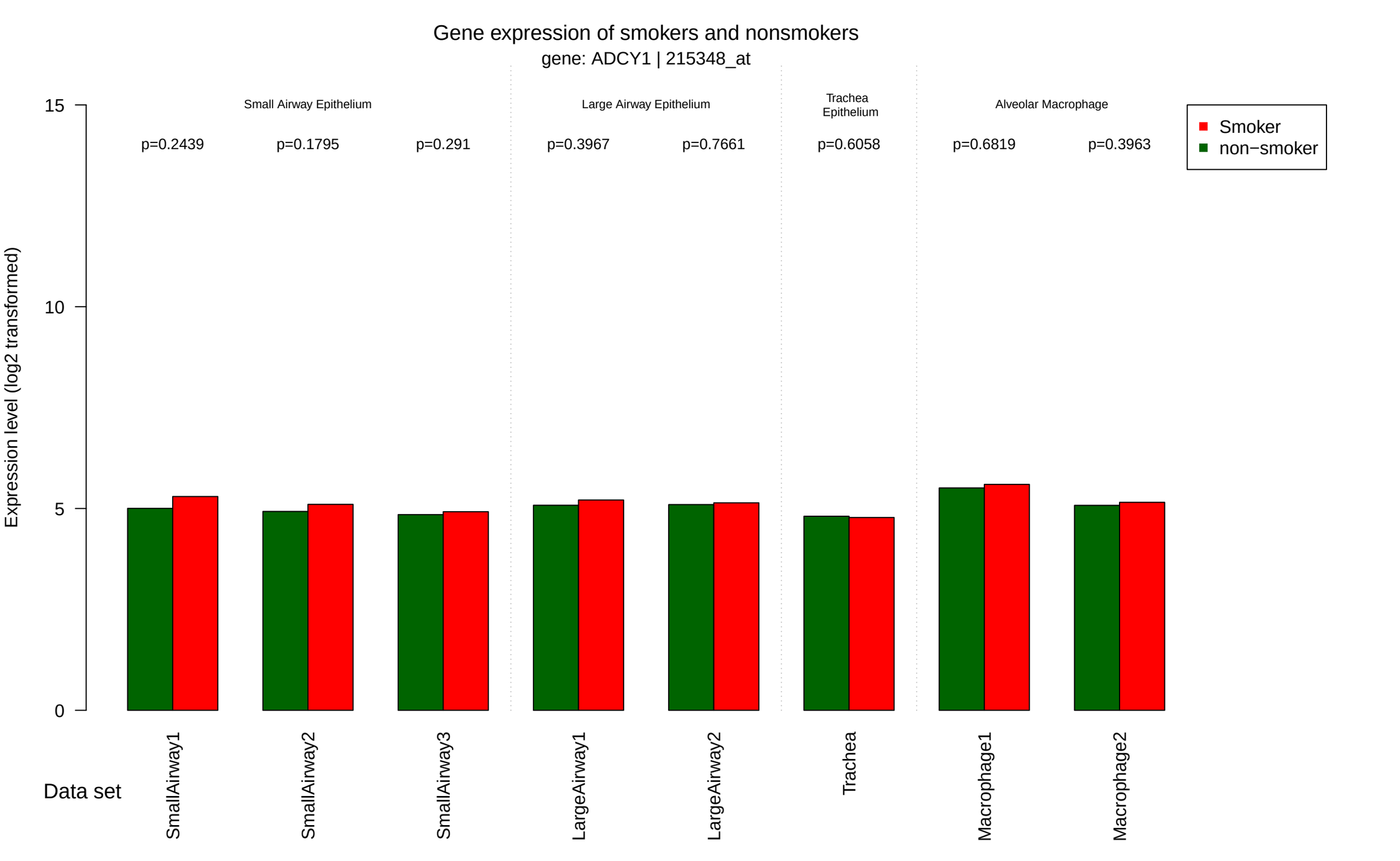
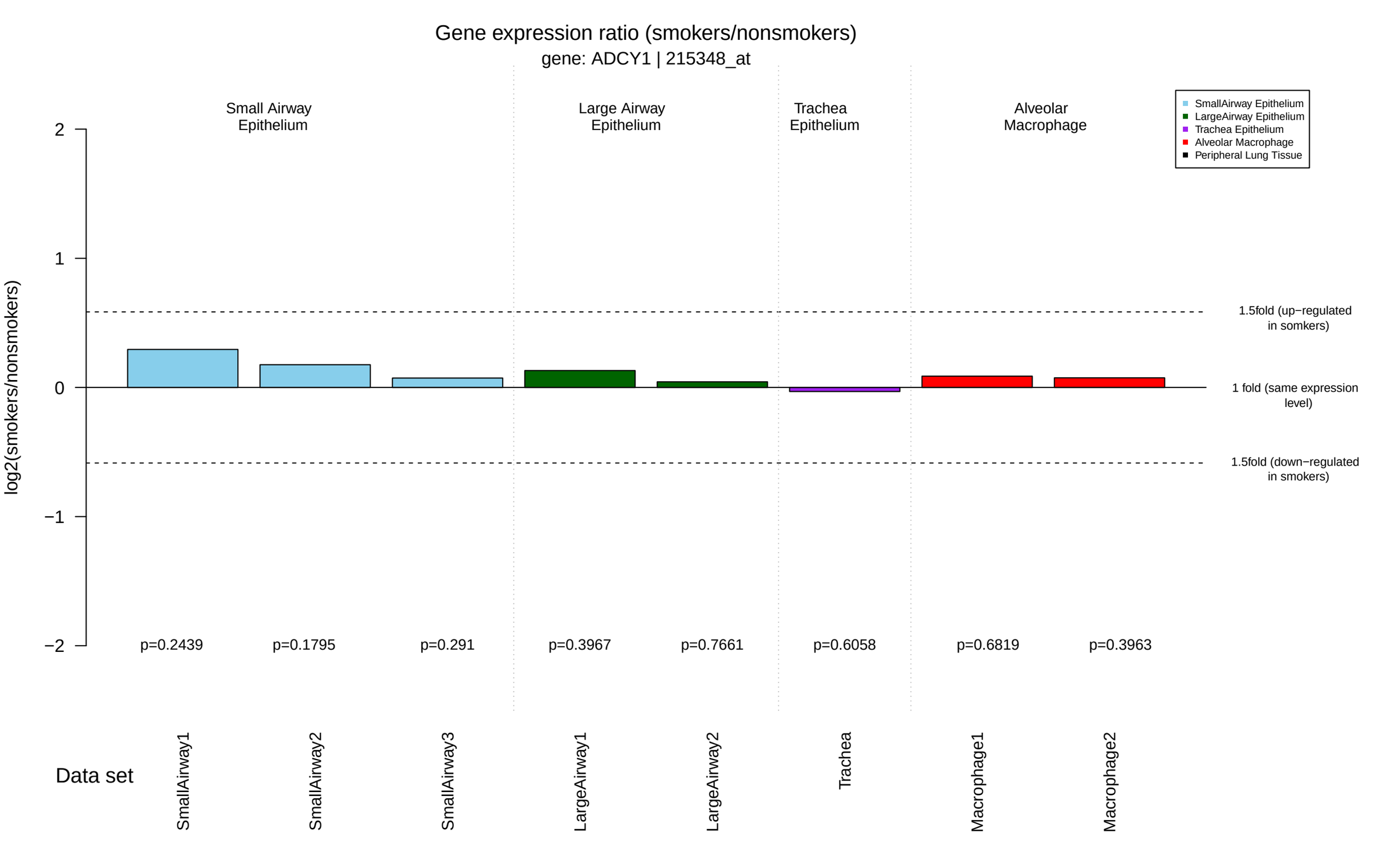
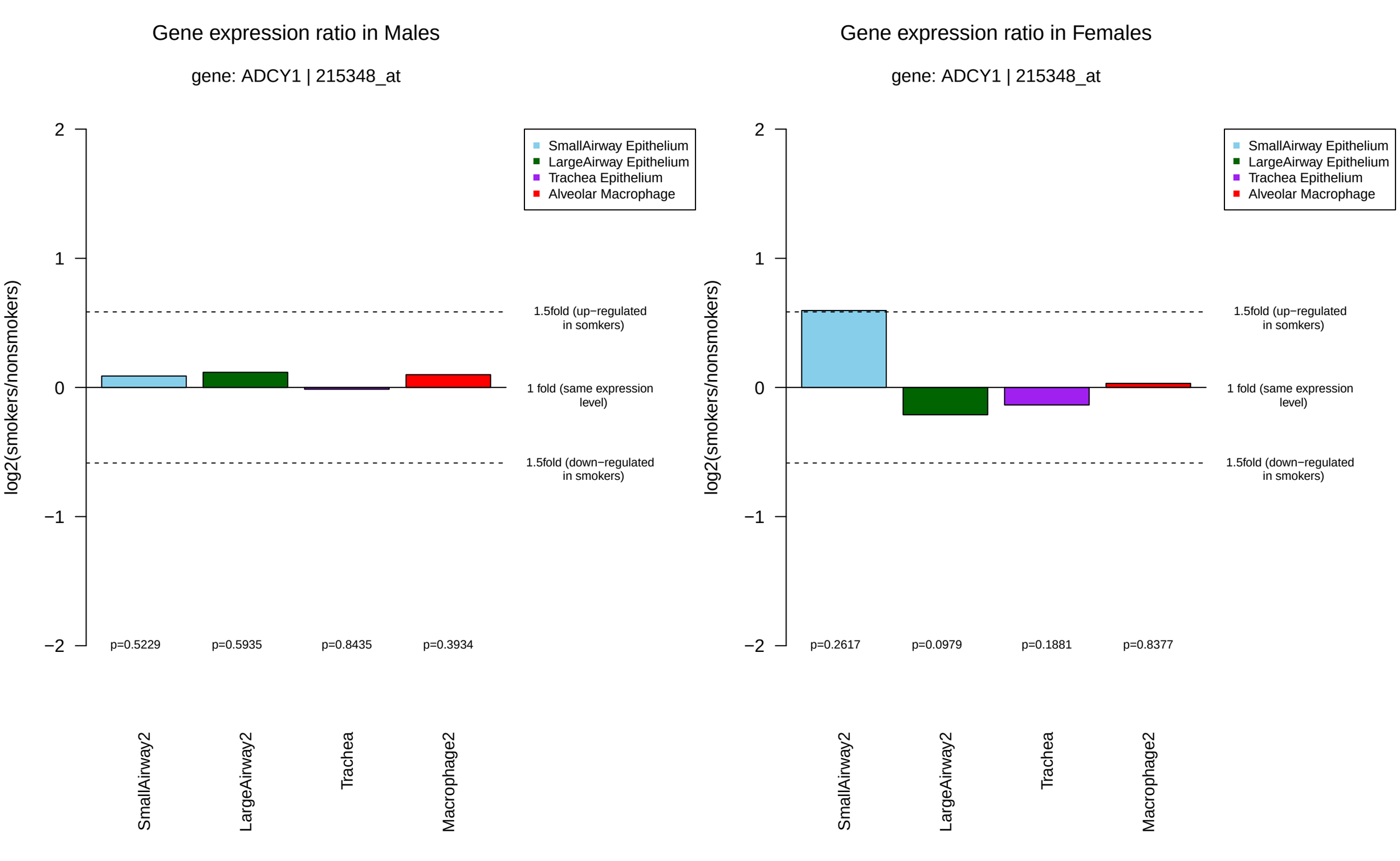
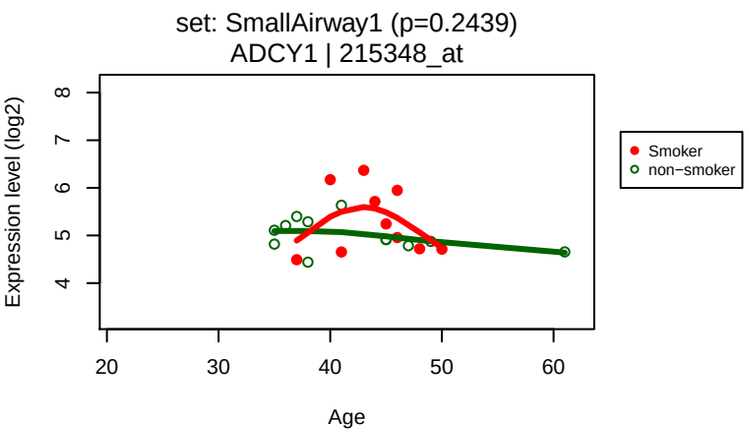
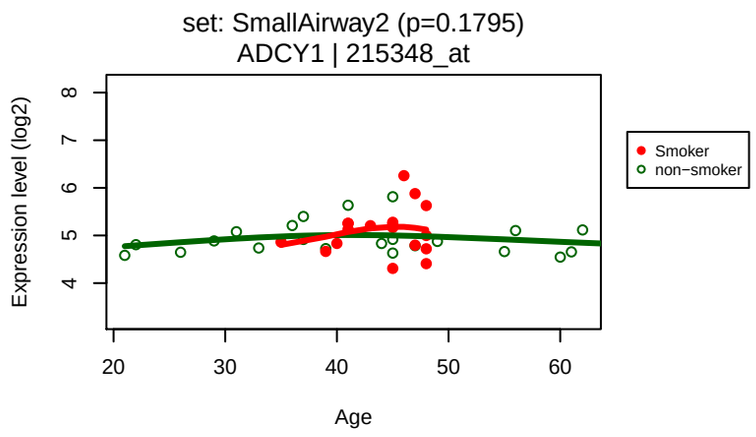
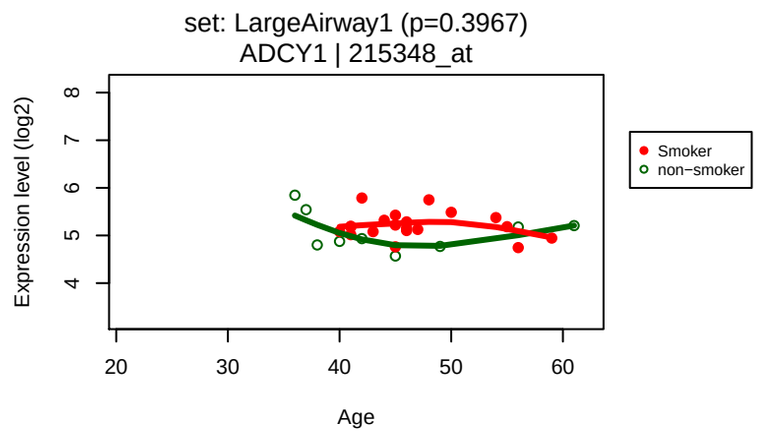

Gene expression of smokers and nonsmokers
gene: ADCY1 | 215348_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2439
p=0.1795
p=0.291
p=0.3967
p=0.7661
p=0.6058
p=0.6819
p=0.3963
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

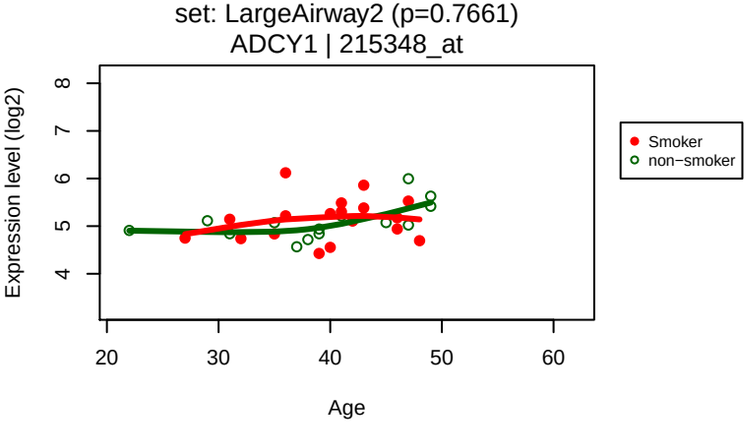

Gene expression ratio (smokers/nonsmokers)
gene: ADCY1 | 215348_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2439
p=0.1795
p=0.291
p=0.3967
p=0.7661
p=0.6058
p=0.6819
p=0.3963
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

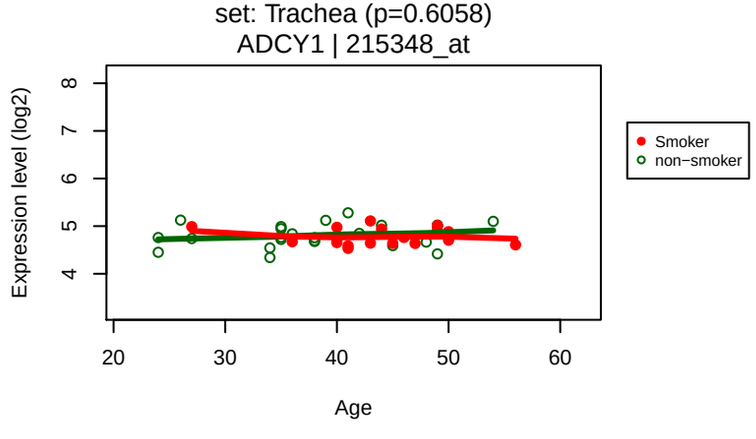

Gene expression ratio in Males
gene: ADCY1 | 215348_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5229
p=0.5935
p=0.8435
p=0.3934
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY1 | 215348_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2617
p=0.0979
p=0.1881
p=0.8377
SmallAirway2
LargeAirway2
Trachea
Macrophage2

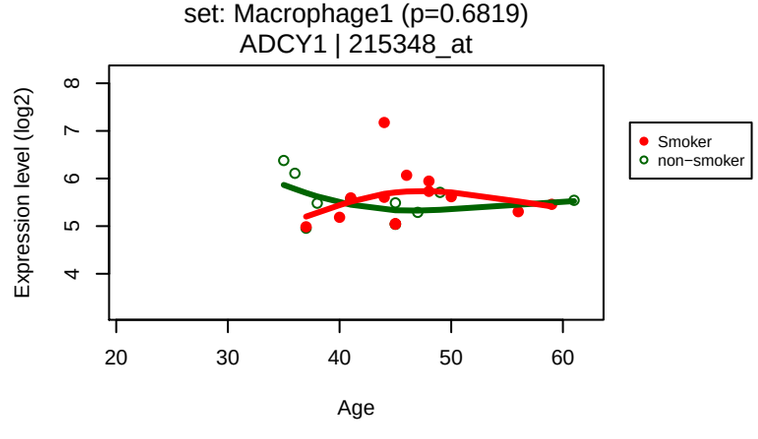

set: SmallAirway1 (p=0.2439)
ADCY1 | 215348_at
Expression level (log2)
Age
Smoker
non−smoker

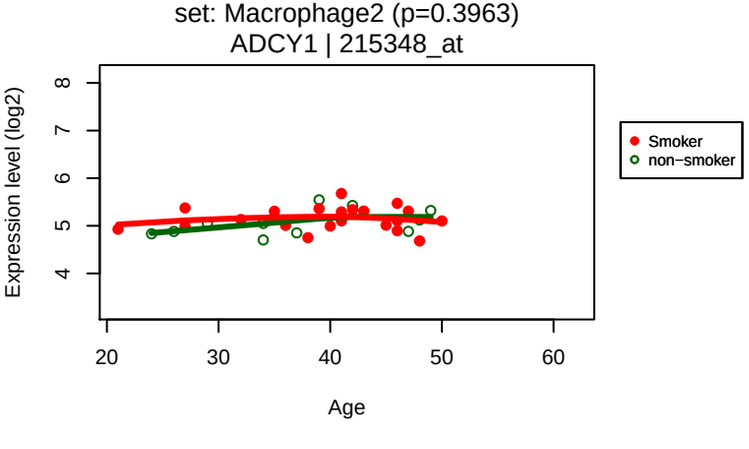

set: SmallAirway2 (p=0.1795)
ADCY1 | 215348_at
Expression level (log2)
Age
Smoker
non−smoker

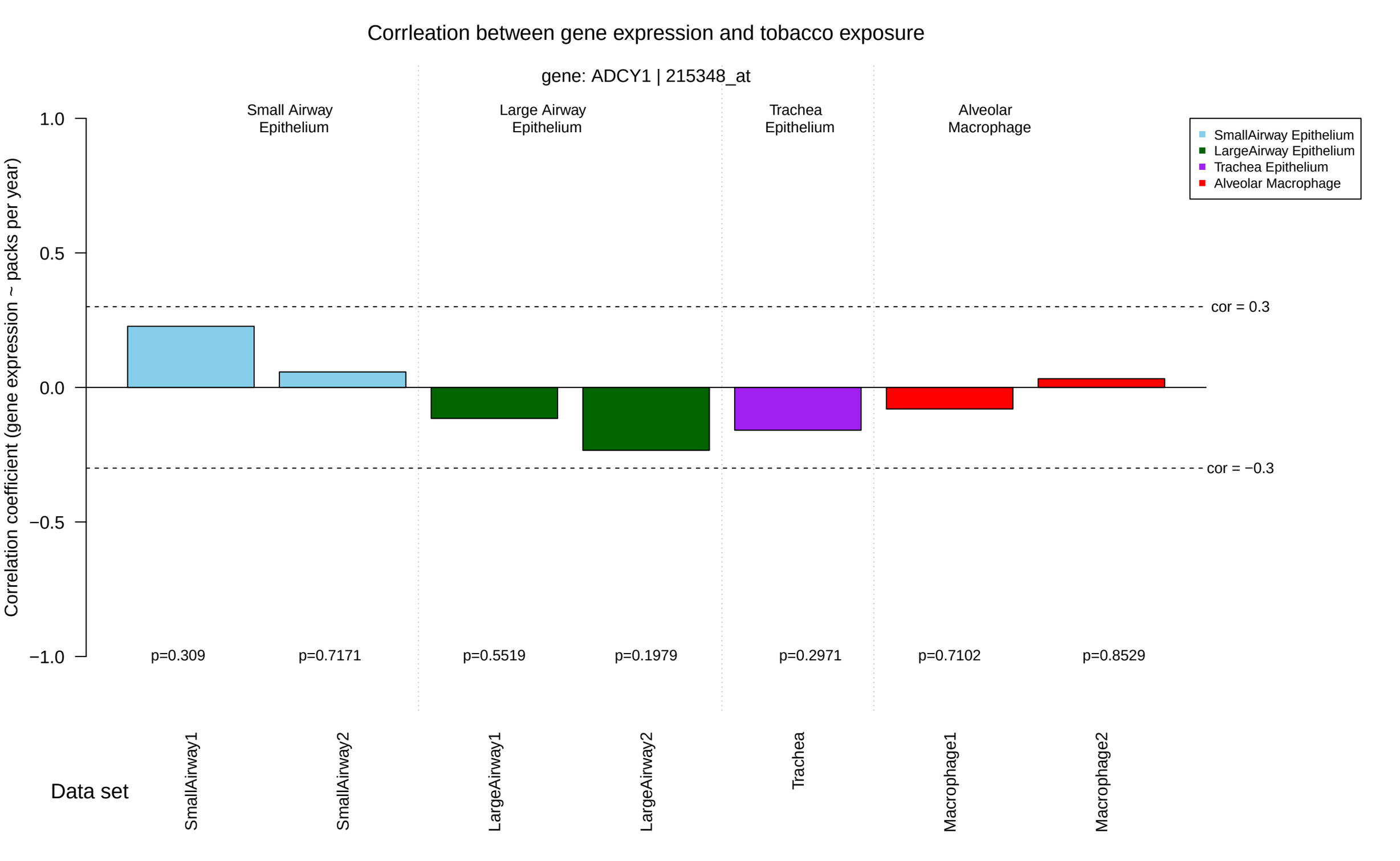

set: LargeAirway1 (p=0.3967)
ADCY1 | 215348_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7661)
ADCY1 | 215348_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6058)
ADCY1 | 215348_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6819)
ADCY1 | 215348_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3963)
ADCY1 | 215348_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY1 | 215348_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.309
p=0.7171
p=0.5519
p=0.1979
p=0.2971
p=0.7102
p=0.8529
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2